DATA STRUCTURE ASSIGNMENT UNIT-1

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1. What is an ADT?

An Abstract Data Type (ADT) is a logical model of a data structure that defines what operations can be performed but not how they are implemented. A data structure is the actual implementation in memory.

For example, Stack is an ADT. It supports push and pop operations. In Python, we implemented Stack using a list.

Stack operations like push and pop are O(1) because they only add or remove elements from the end of the list, which takes constant time.

2. Factorial Analysis

Factorial uses recursion.

Time Complexity:

O(n) - because it makes n recursive calls.

Space Complexity:

O(n) - because recursion depth is n.

Best, Average, Worst Case:

All are O(n) because recursion always runs n times.

3. Fibonacci Analysis

Naive Fibonacci

Time Complexity: O(2")

Space Complexity: O(n)

It is slow because it recalculates the same values many times.

Memoized Fibonacci

Time Complexity: O(n)

Space Complexity: O(n)

Memoization stores previously computed results, avoiding repeated calculations